Subject: Mathematics

Investigating patterns in the product of the sum of divisors function σ and Euler's totient function φ

Research Question: What gives rise to the patterns in the graph of

$f(n) = n^{-2}\sigma(n)\varphi(n)$?

Word count: 1729

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1 Introduction

The theory of numbers plays an important part in the study of mathematics, and it means determining certain properties of positive integers. For example, one might ask, 'Which positive integers may not be written as a product of two smaller positive integers?' We call such numbers *primes*, and already in Ancient Greece, Euclid had proven that there are infinitely many prime numbers. Another mathematician from the time, Nicomachus of Gerasa, had conjectured that there are infinitely many *perfect numbers*, which are numbers that equal the sum of their proper divisors, i.e. positive divisors which are smaller than the number itself. For example, 6 is perfect because $6 = 1 + 2 + 3$, and numbers 1, 2, 3 are exactly the positive integers which are less than 6 and divide 6. Yet, unlike the existence of infinitely many primes, this latter statement has remained unproven for now more than two millenia since its proposal.

To investigate and potentially solve these elusive questions, mathematicians have defined and studied certain functions which report an interesting property of a number. My extended essay will be about investigating the product of two functions which are both central in number theory.

Firstly, we have the function $\sigma(n)$, which is defined as the sum of all positive divisors of n. This is almost directly a part of the definition of perfect numbers, which in turn are connected to an unsolved problem, making the function inherently interesting to mathematicians. The slight difference is that summing over all positive divisors includes the number n itself, alongside all its proper divisors. Therefore, a positive integer n is perfect if $n = \sigma(n) - n$, i.e. it is equal to the sum of all its positive divisors excluding itself.

Next, we have Euler's Totient Function, denoted by $\varphi(n)$, which is defined as the number of positive integers less than or equal to n and have no common divisors with

n. This function has become highly relevant in the past century, because it is used many computer algorithms such as encryption and quickly checking whether a very large number is prime.

My study will focus on the function $f(n) = \frac{\sigma(n)\varphi(n)}{n^2}$, which is the product of these two aformentioned functions, divided by n^2 because (surprisingly) for any positive integer n, the values $\sigma(n)\varphi(n)$ and n^2 actually are comparable in size.

We will first develop the necessary tools to tell how close $\sigma(n)\varphi(n)$ and n^2 really lie, and then investigate the following figure, which is obtained when our function $f(n) = \frac{\sigma(n)\varphi(n)}{n^2}$ is plotted on the Cartesian plane, as we see in the following figure:

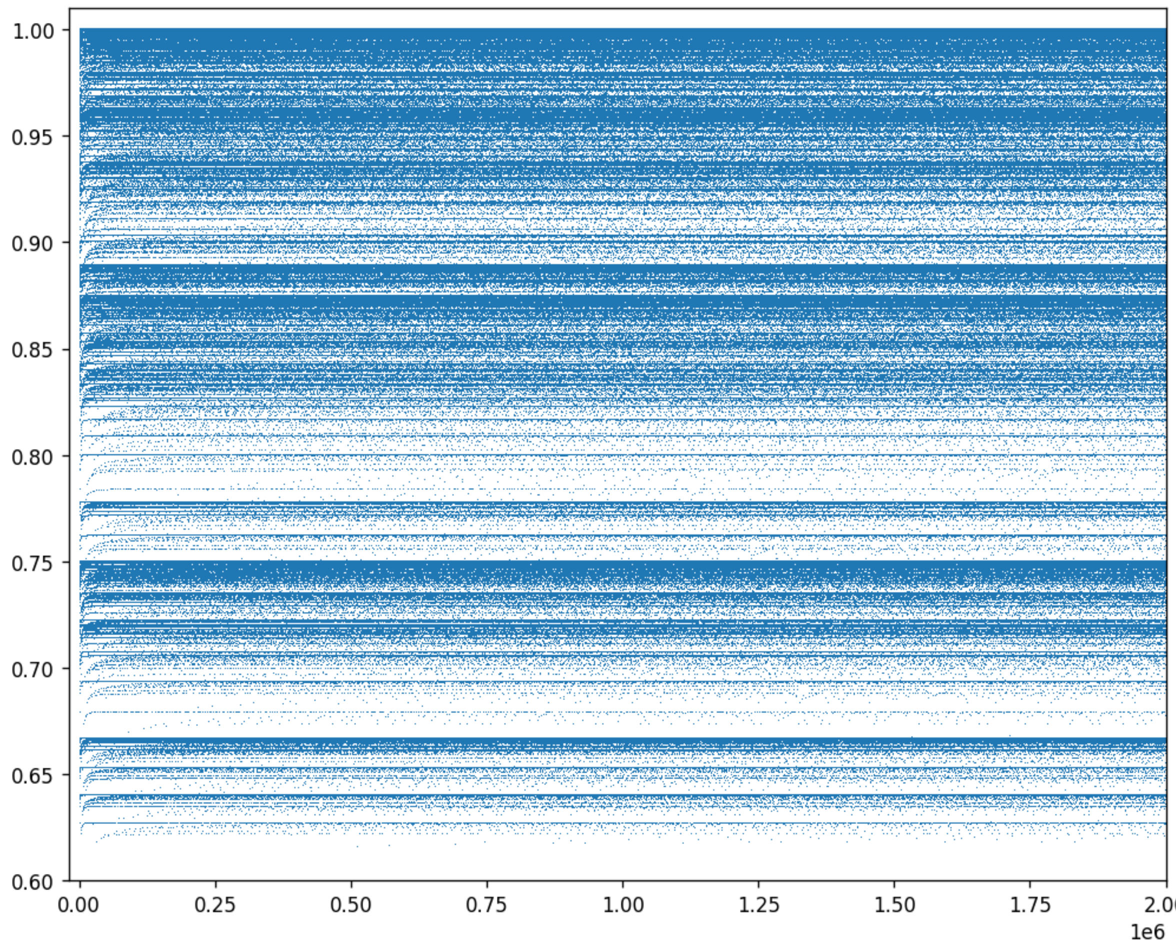

Figure 1: The graph of $f(n)$, up to $n = 2,000,000$

The second half of my investigation aims to explain some properties of Figure 1, of which the most striking are the horizontal lines, which emerge in a seemingly random pattern. Specifically, we will answer the following questions:

Question 1. *Why are 'dense lines' formed?*

Question 2. *Where can 'dense lines' form?*

2 Prerequisite Knowledge, Notation, and Definitions

The inarguably most important theorem in number theory is the following:

Theorem 1 (Fundamental Theorem of Arithmetic)**.** *Every positive integer has a unique prime factorization.*

This theorem is crucial in our ensuing discussion, because it allows us to express any integer $n \geq 2$ uniquely in terms of its k prime divisors $p_1, p_2, ..., p_k$, along with the positive integers $\alpha_1, \alpha_2, ..., \alpha_k$ which indicate the number of times that n is divisible by the corresponding prime factor. With this, we obtain the familiar form

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k} = \prod_{i=1}^{k} p_i^{\alpha_i} \tag{2.1}$$

In the special case $n = 1$, there are no prime factors, and by convention

$$\textit{the empty product equals 1.} \tag{2.2}$$

The reader should be comfortable with Π-notation for sequential products, which is similar to Σ but for multiplication.

2.1 Introduction to Sets

A *set* is an *unordered* collection of elements. The discussion in this paper only involves sets of numbers, such as $\mathbb{Z}$, used to denote the set of integers, and $\mathbb{R}$, used to denote the set of real numbers.

Here *unordered* means that for example, the set $A = \{1, 2\}$ is the same as the set $B = \{2, 1\}$. When two sets A, B are the same, we write $A = B$.

Notation. Let A, B, and S be sets.

- The notation $x \in S$ means that x is an element of the set S.
- For a finite set S, the symbol $|S|$ denotes the number of elements contained in S.

2.2 Set-builder Notation

Set-builder notation is a way for mathematicians to define sets symbolically rather than verbosely. Basic set-builder notation is used frequently throughout this paper.

Notation. The set S of elements in the set A that fulfill a certain condition is defined, i.e. 'built', with the following notation:

$$S = \{a \in A \ : \ \text{conditions for } a\}$$

In this case, S is a subset of A, because every element of S is an element of A. This is denoted by $S \subseteq A$. The reader should understand the following examples before proceeding:

Example 1. Build the set of positive integers, denoted $\mathbb{Z}^+$, by choosing the integers which are positive:

$$\mathbb{Z}^+ = \{x \in \mathbb{Z} \ : \ x > 0\} \tag{2.3}$$

Example 2. Build the set of positive divisors of an integer $n \neq 0$, denoted $\mathcal{D}_n$, by choosing the positive integers that divide n:

$$\mathcal{D}_n = \{x \in \mathbb{Z}^+ \ : \ x \mid n\} \tag{2.4}$$

The notation $a \mid b$ means a divides b.

Example 3. Recall that prime numbers are positive integers greater than 1 whose only positive divisors are 1 and itself. Build the set of primes, denoted $\mathbb{P}$, using the set of positive divisors defined in (2.4):

$$\mathbb{P} = \big\{x \in \mathbb{Z}^+ \ : \ x > 1, \ \mathcal{D}_x = \{1, \ x\}\big\} \tag{2.5}$$

In this case, multiple conditions on x are necessary, and notationally we separate them with a comma.

More complicated sequential sums and products also require set-builder-like notation in the subscripts, used to impose restrictions on what to take the sum or product over. (2.6) in **Definition 1** below serves as an example of this.

2.3 Key Definitions

As alluded in the introduction, the following functions, defined for all positive integers n, are the central focus of this study.

Definition 1 (Sum of Divisors Function σ)**.** Define $\sigma(n)$ as the sum of all positive divisors of n.

Mimicking set-builder notation, we can write

$$\sigma(n) = \sum_{d \in \mathcal{D}_n} d \tag{2.6}$$

Example 4. Compute $\sigma(12)$: Since the positive divisors of 12 are $\mathcal{D}_{12} = \{1, 2, 3, 4, 6, 12\}$, taking their sum yields $\sigma(12) = 1 + 2 + 3 + 4 + 6 + 12 = 28$

Definition 2 (Euler's Totient Function φ)**.** Define $\varphi(n)$ as the number of positive integers less than or equal to n and *coprime* to n. Formally, define

$$\varphi(n) = \left|\{x \in \mathbb{Z}^+ \; : \; x \leq n, \; \gcd(x, n) = 1\}\right| \tag{2.7}$$

Terminology. We say that 'a is coprime to b', or 'a and b are coprime', if $\gcd(a, b) = 1$.

Example 5. Compute $\varphi(12)$:

The positive integers less than or equal to 12 and coprime to 12 are 1, 5, 7, 11. Since there are 4 such numbers, $\varphi(12) = 4$.

3 The Problem of Study

The author first learned of this problem in the undergraduate textbook *Introduction to Analytic Number Theory* [2].

Problem 1. Define $f(n) = \frac{\sigma(n)\varphi(n)}{n^2}$. Show that $\dfrac{6}{\pi^2} < f(n) < 1$ for all integers n greater than 1.

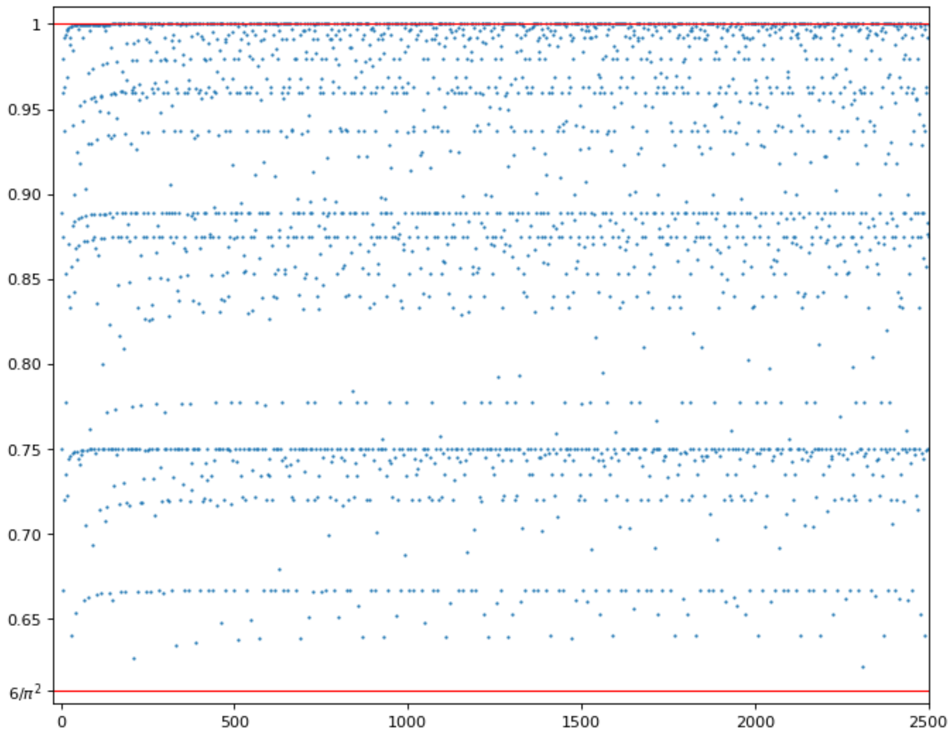

Figure 2: Graph of $f(n)$ up to $n = 2500$, with the proposed bounds in **Problem 1**

Figure 2 visualizes the scenario by plotting points $(n, f(n))$ (in blue) and the lower– and upper bounds (in red) on the Cartesian plane. Then, the task in **Problem 1** is to show that all blue points fall in between the red lines.

Moreover, the 'patterns' that we will later investigate are also apparent in this graph. In addition to solving **Problem 1**, this paper will develop the necessary tools to answer the following questions.

Question 1. *Why are 'dense lines' formed?*

This question regards the horizontal lines that many of the points seem to fall in, at $y = 1$, $y = 0.75$, and $y = 0.89$ for example. This is one of the first questions arising upon seeing **Figure 2**, since the orderly patterns strike as unexpected in such a scattered and seemingly random distribution. After addressing **Question 1**, the next natural question is,

Question 2. *Where can 'dense lines' form?*

This is also a natural thing to ask; we see many dense lines between 0.95 and 1, and much fewer below 0.70. Is it true that the denser lines may only form in certain regions, and are entirely absent elsewhere? As we will show, the answer is that anywhere between $6/\pi^2$ and 1 are denser lines able to emerge. The reader may disregard these questions for now, as an exact definition of these 'dense lines' is not given until later.

4 Background for Problem 1

To approach **Problem 1**, we must improve our understanding on the functions σ and φ. The following product forms for these functions in terms of the prime factorization of n shall be derived.

Lemma 1. *For any positive integer n with prime factorization $n = \prod_{i=1}^{k} p_i^{\alpha_i}$ holds*

$$\sigma(n) = \prod_{i=1}^{k} \frac{p_i^{\alpha_i+1} - 1}{p_i - 1} \tag{4.1}$$

$$\varphi(n) = \prod_{i=1}^{k} p_i^{\alpha_i-1}(p_i - 1) \tag{4.2}$$

4.1 Deriving the Product Form of σ

To do this, we first show that σ is *multiplicative*, which means

$$\sigma(mn) = \sigma(m)\sigma(n) \tag{4.3}$$

holds for all coprime positive integers m, n. The following lemma is used to show *multiplicativity*.

Lemma 2. *For coprime positive integers m, n, each element d in the set $\mathcal{D}_{mn}$ is a product of a unique pair of elements in $\mathcal{D}_m$ and $\mathcal{D}_n$.*

Conversely, each pair of elements in $\mathcal{D}_m$ and $\mathcal{D}_n$ has a product which is a unique element in $\mathcal{D}_{mn}$.

Example 6. Let $m = 9$, $n = 14$. The divisors of 9 are $\mathcal{D}_9 = \{1, 3, 9\}$ and of 14 are $\mathcal{D}_{14} = \{1, 2, 7, 14\}$. Every divisor of $mn = 126$ is a product of one divisor of 9 and one divisor of 14:

$$\begin{aligned}
1 &= 1 \cdot 1, & 2 &= 1 \cdot 2, & 7 &= 1 \cdot 7, & 14 &= 1 \cdot 14,\\
3 &= 3 \cdot 1, & 6 &= 3 \cdot 2, & 21 &= 3 \cdot 7, & 42 &= 3 \cdot 14,\\
9 &= 9 \cdot 1, & 18 &= 9 \cdot 2, & 63 &= 9 \cdot 7, & 126 &= 9 \cdot 14.
\end{aligned}$$

Proof of Lemma 2

Let d be a positive divisor of mn, and write the prime factorization of d:

$$d = \prod_{i=1}^{k} p_i^{\alpha_i} \tag{4.4}$$

In this form, each term $p_i^{\alpha_i}$ either

1. divides m and is coprime to n, or

2. divides n and is coprime to m,

because d is a divisor of mn and m, n are coprime. This means we can write

$$d = d_m d_n \tag{4.5}$$

where d_m is the product of the terms which correspond to case 1, and d_n is the product of the terms which correspond to case 2. In this way, we guarantee that $d_m \mid m$ and $d_n \mid n$, or equivalently $d_m \in \mathcal{D}_m$ and $d_n \in \mathcal{D}_n$.

This shows that each element in $\mathcal{D}_{mn}$ is a product of some pair of elements $d_m \in \mathcal{D}_m$ and $d_n \in \mathcal{D}_n$. The process of separating the terms $p_i^{\alpha_i}$ between case 1 and case 2 does not involve choices, so the corresponding pair (d_m, d_n) is unique.

Conversely, every pair $d_m \in \mathcal{D}_m$, $d_n \in \mathcal{D}_n$ determines exactly one divisor of mn, namely their product $d_m d_n$. $\square$

Proving multiplicativity of σ

Using **Lemma 2**, *multiplicativity* can be established using **Definition 1**.

$$\begin{aligned}
\sigma(mn) &= \sum_{d \in \mathcal{D}_{mn}} d && (4.6)\\
&= \sum_{d_m \in \mathcal{D}_m} \sum_{d_n \in \mathcal{D}_n} d_m d_n && (4.7)\\
&= \sum_{d_m \in \mathcal{D}_m} d_m \left(\sum_{d_n \in \mathcal{D}_n} d_n \right) && (4.8)\\
&= \left(\sum_{d_m \in \mathcal{D}_m} d_m \right) \left(\sum_{d_n \in \mathcal{D}_n} d_n \right) = \sigma(m)\sigma(n). && (4.9)
\end{aligned}$$

(4.6) This is **Definition 1**.

(4.7) By **Lemma 2**, divisors of mn correspond uniquely to products $d_m d_n$ with $d_m \mid m$, $d_n \mid n$, so the sum may be rewritten as a double sum over such pairs.

(4.8) The inner and outer sums are independent, allowing the factors d_m and d_n to be separated.

(4.9) Since the inner sum depends only on n, it can be factored out of the outer sum.

Finishing the product form of σ

Multiplicativity lets us express $\sigma(n)$ in terms of the prime factorization $n = \prod_{i=1}^{k} p_i^{\alpha_i}$. Since $p_1^{\alpha_1}, p_2^{\alpha_2}, \ldots, p_k^{\alpha_k}$ are all coprime, we may separate the terms $p_1^{\alpha_1}, p_2^{\alpha_2}, \ldots, p_k^{\alpha_k}$ out one by one.

$$\begin{aligned}
\sigma(n) &= \sigma(p_1^{\alpha_1} p_2^{\alpha_2} \ldots p_k^{\alpha_k}) && (4.10)\\
&= \sigma(p_1^{\alpha_1}) \cdot \sigma(p_2^{\alpha_2} \ldots p_k^{\alpha_k}) && (4.11)\\
&= \sigma(p_1^{\alpha_1}) \cdot \sigma(p_2^{\alpha_2}) \cdot \sigma(p_3^{\alpha_3} \ldots p_k^{\alpha_k}) && (4.12)\\
&= \ldots && (4.13)\\
&= \sigma(p_1^{\alpha_1}) \cdot \sigma(p_2^{\alpha_2}) \cdot \cdots \cdot \sigma(p_k^{\alpha_k}) && (4.14)\\
&= \prod_{i=1}^{k} \sigma(p_i^{\alpha_i}) && (4.15)
\end{aligned}$$

To obtain the desired form in (4.1), we only need to compute $\sigma(p^\alpha)$ when p is prime and α is any positive integer. This is easy since

$$\mathcal{D}_{p^\alpha} = \{1, p, p^2, \ldots, p^\alpha\} \tag{4.16}$$

which implies $\sigma(p^\alpha)$ is simply a geometric series

$$\sigma(p^\alpha) = 1 + p + p^2 + \cdots + p^\alpha = \frac{p^{\alpha+1} - 1}{p - 1} \tag{4.17}$$

Finally, substitute this into (4.15) to finish:

$$\sigma(n) = \prod_{i=1}^{k} \frac{p_i^{\alpha_i+1} - 1}{p_i - 1} \tag{4.1}$$

$\square$

Example 7. Compute $\sigma(2025)$.

$$\begin{aligned}\sigma(2025) =& \sigma(3^4 \cdot 5^2) \\ =& \frac{3^5 - 1}{3 - 1}\frac{5^3 - 1}{5 - 1} \\ =& 121 \cdot 31 = 3751\end{aligned}$$

4.2 Deriving the Product Form of φ

To do this, we first show that φ is *multiplicative*, which means

$$\varphi(mn) = \varphi(m)\varphi(n) \tag{4.18}$$

holds for all coprime positive integers m, n. The following lemma, which is a special case of the **Chinese Remainder Theorem**, is used to show *multiplicativity.*

Lemma 3. *For coprime positive integers m, n, each integer x with $0 \leq x < mn$ has a unique pair of remainders when divided by m and by n.*

Conversely, each pair of remainders is given by a unique integer within this range.

Example 8. Let $m = 3,\ n = 4$. Then $mn = 12$, so we list all integers x with $0 \leq x < 12$ together with their remainders when divided by 3 and by 4:

$$\begin{array}{cccc} 0 \to (0,0), & 3 \to (0,3), & 6 \to (0,2), & 9 \to (0,1), \\ 1 \to (1,1), & 4 \to (1,0), & 7 \to (1,3), & 10 \to (1,2), \\ 2 \to (2,2), & 5 \to (2,1), & 8 \to (2,0), & 11 \to (2,3). \end{array}$$

Proof of Lemma 3

We first show, by contradiction, that each pair of remainders when dividing by m and by n is produced by at most one integer x with $0 \leq x < mn$. Let the remainders of x when divided by m and by n be x_m and x_n, respectively. Assume, for contradiction, that there exists another integer x', with $0 \leq x' < mn$, which also gives remainders x_m, x_n. Without loss of generality, assume $x' > x$.*

Since x and x' have the same remainders when divided by m, we have $m \mid x' - x$. Similarly, $n \mid x' - x$. Recall that m and n are coprime, these can be combined into $mn \mid x' - x$.

But $x' > x$ by assumption, so $x' - x$ is a positive multiple of mn, which implies

$$x' - x \geq mn \tag{4.19}$$

$$x' \geq mn + x \geq mn \tag{4.20}$$

contradicting with the assumption that $0 \leq x' \leq mn$. Therefore, each pair of remainders is be given by at most one integer x with $0 \leq x < mn$.

*This is allowed because the other case is symmetrical

It is known that there are m possible remainders when dividing by m, and n possible remainders when dividing by n. Therefore, there are mn total pairs of remainders. This is equal to the number of integers x that satisfy $0 \leq x < mn$.

Since each such x gives exactly one pair of remainders, and each pair of remainders is given by at most one such x, it is necessarily true that each pair of remainders is given by some such x. $\square$

But $\varphi(n)$ only concerns integers that are coprime to n. The following lemma is necessary.

Lemma 4. *Given coprime positive integers m, n and integer x satisfying $0 \leq x < mn$, let the remainders of x be x_m and x_n when divided by m and by n, respectively.*

Then x is coprime to mn if and only if x_m is coprime to m and x_n is coprime to n. Symbolically, the claim is

$$\gcd(x, mn) = 1 \iff \gcd(x_m, m) = \gcd(x_n, n) = 1 \tag{4.21}$$

Example 9. Look back at **Example 8**: The integers $1, 5, 7, 11$ are coprime to 12. In the corresponding pairs of remainders, the left remainder is coprime to 3, and the right remainder is coprime to 4.

Conversely, for the remaining integers $0, 2, 3, 4, 6, 8, 9, 10$, either the left remainder shares a prime factor with 3, or the right remainder shares a prime factor with 4.

Proof of Lemma 4

Consider the integer division of x by m and by n:

$$x = q_m m + x_m \tag{4.22}$$

$$x = q_n n + x_n \tag{4.23}$$

Here q_m and q_n are the quotients of division by m and by n, respectively.

If $\gcd(x, mn) > 1$, then there exists some prime p which divides x and divides either m or n. Without loss of generality, assume $p \mid m$. Then rearrange (4.22):

$$x - q_m m = x_m \tag{4.24}$$

Both terms in the left hand side are divisible by p, so x_m must be divisible by p as well. Hence p is a common divisor of x_m and m, implying that $\gcd(x_m, m) > 1$.

Conversely, again without loss of generality, assume that $\gcd(x_m, m) > 1$. Then there exists some prime p which divides both x_m and m. Since mn is a multiple of m, it must be divisible by p. Both terms in the right hand side of (4.22) are divisible by p, so x must be divisible by p as well. Hence p is a common divisor of x and mn, implying that $\gcd(x, mn) > 1$.

We have shown that $\gcd(x, mn) > 1$ holds exactly when at least one of $\gcd(x_m, m)$ and $\gcd(x_n, n)$ is greater than 1. Equivalently, $\gcd(x, mn) = 1$ holds if and only if $\gcd(x_m, m) = \gcd(x_n, n) = 1$. $\square$

Proving multiplicativity of φ

Recall that **Definition 2** defines $\varphi(n)$ as the number of positive integers $x \leq n$ which are coprime to n, while **Lemma 3** and **Lemma 4** are concerned with nonnegative integers $x < n$. To address this disparity, we simply note that for any integer $n > 1$, neither n nor 0 are coprime to n, so $\varphi(n)$ is equal to the number of nonnegative $x < n$ which are coprime to n.

Since *multiplicativity* is trivial for $m = 1$ or $n = 1$, it is only necessary to consider coprime integers m, n greater than 1.

Combining **Lemma 3** and **Lemma 4** shows that the total number of pairs of remainders (x_m, x_n) where $\gcd(x_m, m) = \gcd(x_n, n) = 1$ is equal to the number of integers x satisfying $0 \leq x < mn$ and $\gcd(x, mn) = 1$. Note that this is precisely the amended definition of $\varphi(mn)$.

Moreover, since the possible remainders when dividing by m are simply $0, 1, \ldots, m-1$ and only those coprime to m are counted, the number of possible values for x_m is $\varphi(m)$, again using the amended definition. Similarly, the number of possible values for x_n is $\varphi(n)$.

This means the number of pairs (x_m, x_n) is $\varphi(m)\varphi(n)$. Therefore, $\varphi(mn) = \varphi(m)\varphi(n)$.

Finishing the product form of σ

Now, exactly like (4.10) – (4.15) in the derivation of the product form of σ, we deduce that for $n = \prod_{i=1}^{k} p_i^{\alpha_i}$ holds

$$\varphi(n) = \prod_{i=1}^{k} \varphi(p_i^{\alpha_i}) \tag{4.25}$$

For any prime p and any positive integer α, we may compute $\varphi(p^\alpha)$ by counting the numbers from 1 to p^α and removing all $p^{\alpha-1}$ multiples of p:

$$\varphi(p^\alpha) = p^\alpha - p^{\alpha-1} = p^{\alpha-1}(p-1) \tag{4.26}$$

Finally, substitute this into (4.25) to finish:

$$\varphi(n) = \prod_{i=1}^{k} p_i^{\alpha_i - 1}(p_i - 1) \tag{4.2}$$

$\square$

Example 10. Compute $\varphi(2025)$.

$$\begin{aligned}\varphi(2025) =& \varphi(3^4 \cdot 5^2)\\ =& 3^(4-1)(3-1) \cdot 5^(2-1)(5-1)\\ =& 54 \cdot 20 = 1080\end{aligned}$$

5 Solving Problem 1

'Why is pi here? And why is it squared?' – Grant Sanderson

The task is to prove $\frac{6}{\pi^2} < f(n) < 1$ for all integers $n > 1$. Now that both parts of Lemma 1 have been derived, we can derive the following formula for $f(n) = \frac{\sigma(n)\varphi(n)}{n^2}$ in terms of the prime factorization of n.

Lemma 5. *For any positive integer n with prime factorization $n = \prod_{i=1}^{k} p_i^{\alpha_i}$ holds*

$$f(n) = \prod_{i=1}^{k} \frac{p_i^{\alpha_i+1} - 1}{p_i^{\alpha_i+1}} \tag{5.1}$$

Proof of Lemma 5

Suppose that the integer $n > 1$ has prime factorization $n = \prod_{i=1}^{k} p_i^{\alpha_i}$. By (4.1) and (4.2),

$$\sigma(n)\varphi(n) = \left(\prod_{i=1}^{k} \frac{p_i^{\alpha_i+1}-1}{p_i-1}\right)\left(\prod_{i=1}^{k} p_i^{\alpha_i-1}(p_i-1)\right) \tag{5.2}$$

$$= \prod_{i=1}^{k} \frac{p_i^{\alpha_i+1}-1}{\cancel{p_i-1}} p_i^{\alpha_i-1}(\cancel{p_i-1}) \tag{5.3}$$

$$= \prod_{i=1}^{k} (p_i^{\alpha_i+1}-1)p_i^{\alpha_i-1} \tag{5.4}$$

Additionally,

$$n^{-2} = \left(\prod_{i=1}^{k} p_i^{\alpha_i}\right)^{-2} \tag{5.5}$$

$$= \prod_{i=1}^{k} (p_i^{\alpha_i})^{-2} \tag{5.6}$$

$$= \prod_{i=1}^{k} p_i^{-2\alpha_i} \tag{5.7}$$

Combine (5.4) and (5.7) and simplify:

$$f(n) = \sigma(n)\varphi(n)n^{-2} = \left(\prod_{i=1}^{k} (p_i^{\alpha_i+1}-1)p_i^{\alpha_i-1}\right)\left(\prod_{i=1}^{k} p_i^{-2\alpha_i}\right) \tag{5.8}$$

$$= \prod_{i=1}^{k} (p_i^{\alpha_i+1}-1)p_i^{\alpha_i-1}p_i^{-2\alpha_i} \tag{5.9}$$

$$= \prod_{i=1}^{k} (p_i^{\alpha_i+1}-1)p_i^{\alpha_i-1-2\alpha_i} \tag{5.10}$$

$$= \prod_{i=1}^{k} (p_i^{\alpha_i+1}-1)p_i^{-\alpha_i-1} \tag{5.11}$$

$$= \prod_{i=1}^{k} \frac{p_i^{\alpha_i+1}-1}{p_i^{\alpha_i+1}} \tag{5.12}$$

$\square$

Example 11. Compute $f(2025)$.

Since $2025 = 3^4 \cdot 5^2$, we use (5.1) in **Lemma 5**

$$f(2025) = \frac{(3^{4+1} - 1)}{3^{4+1}} \frac{(5^{2+1} - 1)}{5^{2+1}} = \frac{242}{243} \cdot \frac{124}{125} = \frac{30008}{30375}$$

Verifying using the answers of **7** and **10**:

$$f(2025) = \sigma(2025)\varphi(2025)2025^{-2} = 3751 \cdot 1080 \cdot 2025^-2 = \frac{4051080}{4100625} = \frac{30008}{30375}$$

The value lies between $\frac{\pi^2}{6}$ and 1, as expected.

5.1 Upper bound

Actually, **Lemma 5** looks very promising in regards to solving **Problem 1**. In fact, the upper bound $\sigma(n)\varphi(n)n^{-2} < 1$ is now obvious: Every term in the product is strictly less than 1, so the whole product must be less than 1 whenever the product contains at least one term. That is, whenever $n > 1$.

5.2 Lower bound

The result $f(n) > \frac{6}{\pi^2}$ can be derived using the following lemmas:

Lemma 6. *For any positive integer n with prime factorization $n = \prod_{i=1}^{k} p_i^{\alpha_i}$, we have*

$$f(n) = \prod_{i=1}^{k} \frac{p_i^{\alpha_i+1} - 1}{p_i^{\alpha_i+1}} \geq \prod_{i=1}^{k} \frac{p_i^2 - 1}{p_i^2} \tag{5.13}$$

with equality when $\alpha_1 = \alpha_2 = \cdots = \alpha_k = 1$.

Lemma 7 (Basel Problem)**.** *The infinite product over all primes* $\displaystyle\prod_{p\ prime} \frac{p^2}{p^2-1}$ *is equal to the infinite sum* $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$ *and evaluates to* $\dfrac{\pi^2}{6}$

Proof of Lemma 6

We compare the left product and the right product of (5.13) term by term. For any i holds

$$\frac{p_i^{\alpha_i+1}-1}{p_i^{\alpha_i+1}} = 1 - \frac{1}{p_i^{\alpha_i+1}} \geq 1 - \frac{1}{p_i^{1+1}} = 1 - \frac{1}{p_i^2} = \frac{p_i^2-1}{p_i^2} \tag{5.14}$$

because α_i is a positive integer. Moreover, there is equality in (5.14) when $\alpha_i = 1$.

This means every term in the left product is greater or equal to the corresponding term in the right product, so the whole left product must be greater than or equal to the whole right product. Equality holds when all terms are equal, so $\alpha_i = 1$ for all i. $\square$

Example 12. Bound $f(2025)$ from below.

The prime factorization is $2025 = 3^4 5^2$. Therefore, by **Lemma 6**,

$$f(2025) \geq \frac{3^2-1}{3^2} \cdot \frac{5^2-1}{5^2} = \frac{8}{9} \cdot \frac{24}{25} = \frac{64}{75}$$

Proof of Lemma 7

The author will take for granted that the infinite sum $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$, and a proof will be omitted. For the interested reader, there is a purely elementary geometrical proof[4] popularised by the internet mathematician 3Blue1Brown[3]. A much shorter proof involving integration[1] by the author of *Introduction to Analytic Number Theory* is also recommended.

This paper will only show that the infinite product over all primes and the infinite sum over positive integers are equal.

$$\begin{aligned}
\prod_{p \text{ prime}} \frac{p^2}{p^2-1} &= \prod_{p \text{ prime}} \frac{1}{1-\frac{1}{p^2}} && (5.15)\\
&= \prod_{p \text{ prime}} \left(\sum_{i=0}^{\infty} \left(\frac{1}{p^2} \right)^i \right) && (5.16)\\
&= \prod_{p \text{ prime}} \left(1 + \frac{1}{p^2} + \frac{1}{p^4} + \dots \right) && (5.17)\\
&= \left(1 + \frac{1}{2^2} + \frac{1}{2^4} + \dots \right) \left(1 + \frac{1}{3^2} + \frac{1}{3^4} + \dots \right) \dots && (5.18)\\
&= 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} \dots = \sum_{n=1}^{\infty} \frac{1}{n^2} && (5.19)
\end{aligned}$$

(5.15) Divide the numerator and denominator by p^2.

(5.16) Expand $\frac{1}{1-1/p^2}$ into a geometric series.

(5.17) Write without Σ-notation.

(5.18) Write without Π-notation.

The step between (5.18) and (5.19) feels like a jump. But each term in the infinite sum in (5.19) is produced exactly once when the infinite product in (5.19) is expanded. For example, the term $\frac{1}{4^2}$ arises from selecting $\frac{1}{2^4}$ in the factor corresponding to $p = 2$ and 1 in all the remaining factors. Similarly, the term $\frac{1}{6^2}$ arises from selecting $\frac{1}{2^2}$ in the factor for $p = 2$, $\frac{1}{3^2}$ in the factor for $p = 3$, and 1 in all others. In general, each term $\frac{1}{n^2}$ is produced exactly once upon expanding the product, and it corresponds to the unique prime factorisation of n, by the Fundamental Theorem of Arithmetic.

Therefore,

$$\prod_{p \text{ prime}} \frac{p^2}{p^2 - 1} = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}. \tag{5.20}$$

Finishing the lower bound

By **Lemma 6**, for $n = \prod_{i=1}^{k} p_i^{\alpha_i}$ holds

$$f(n) \geq \prod_{i=1}^{k} \frac{p_i^2 - 1}{p_i^2} \tag{5.21}$$

Each factor $\dfrac{p_i^2 - 1}{p_i^2}$ is less than 1, so adding more such factors (i.e. extending the product to more primes) strictly decreases the value. Hence

$$\prod_{i=1}^{k} \frac{p_i^2 - 1}{p_i^2} > \prod_{p \text{ prime}} \frac{p^2 - 1}{p^2} = \left(\prod_{p \text{ prime}} \frac{p^2}{p^2 - 1} \right)^{-1} \tag{5.22}$$

The equality is strict because the left product has finitely many terms, By **Lemma 7** the product on the right equals $(\frac{\pi^2}{6})^{-1} = \frac{6}{\pi^2}$. Therefore,

$$f(n) > \frac{6}{\pi^2}. \tag{5.23}$$

□

6 Dense lines

Now we turn our attention from the bounds of $f(n)$ to the patterns formed in its graph.

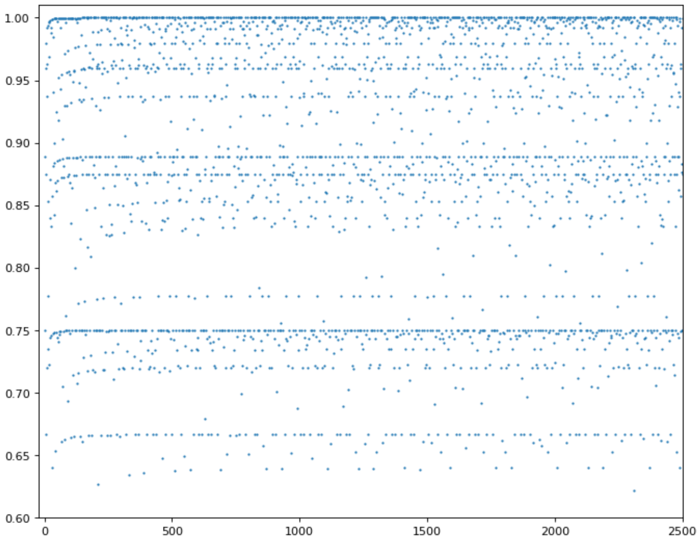

Figure 3: Graph of $f(n)$

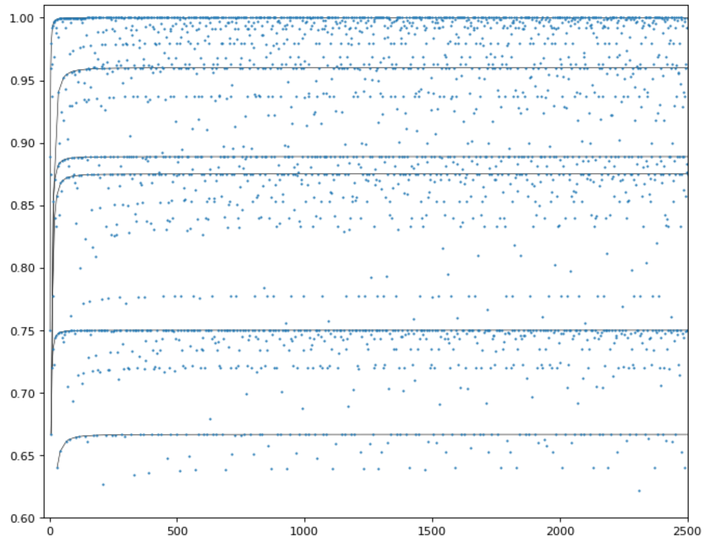

Figure 4: First six 'dense lines'

Recall that 'dense lines' still lacked a precise definition. Therefore, let us attempt to provide one here.

Definition 3. Let $p_1, p_2, \ldots$ be the sequence of prime numbers in ascending order. We say that an infinite sequence of numbers $a_1, a_2, \ldots$ form a 'dense line' if there exist positive real constants M, c, k such that for all terms a_i of the sequence, the following conditions are met.

$$\frac{a_i}{p_i} \leq M \tag{6.1}$$

$$|f(a_i) - k| \leq \frac{c}{a_i^2} \tag{6.2}$$

Condition (6.1) requires that the sequence (a_i) does not grow significantly faster than the sequence of prime numbers (p_i). In particular, it ensures that the ratio $\frac{a_i}{p_i}$ remains bounded by some constant M. If (a_i) grows too rapidly, no such M can exist, and the sequence cannot be considered dense with respect to the primes.

We interpret the minimal constant M satisfying (6.1) for a given sequence (a_i) as a meausre of its density relative to the primes. A smaller M therefore corresponds to a denser sequence.

Condition (6.2) ensures that the points $(a_i, f(a_i))$ asymptotically lie close to a horizontal line. Because the bound $\frac{c}{a_i^2}$ decreases rapidly to zero with increasing a_i, this condition guarantees that the deviation of $f(a_i)$ from k diminishes along the sequence, so that $(f(a_i))$ converges rapidly toward k.

We interpret the minimal constant c satisfying (6.2) for a given sequence (a_i) as a measure of 'how quickly the points fall onto the line.' A smaller c therefore corresponds to a sequence that approaches the line faster.

Example 13. Let $a_i = p_i^2$, where p_i denotes the i-th prime number. Then

$$\frac{a_i}{p_i} = p_i$$

which tends to infinity as i increases, so no such M may exist. Hence, the sequence (a_i) fails to satisfy (6.1), so it is not a dense line.

Example 14 (The primes form a dense line at $t = 1$)**.** Let $a_i := p_i$ be the i-th prime. We claim that $(p_i)_{i\geq 1}$ is a dense line at $y = 1$.

Proof. **(6.1)** For each $x > 0$ the set $\{a_i : a_i \leq x\}$ is exactly the set of primes $\{p_i : p_i \leq x\}$. Hence

$$\frac{\left|\{a_i : a_i \leq x\}\right|}{\left|\{p_i : p_i \leq x\}\right|} = \frac{\left|\{p_i : p_i \leq x\}\right|}{\left|\{p_i : p_i \leq x\}\right|} = 1.$$

The limit as $x \to \infty$ therefore exists and equals $1 > 0$, so condition (1) is satisfied.

(6.2) Recall the definition of f:

$$f(n) = \frac{\sigma(n)\,\varphi(n)}{n^2}.$$

If p is prime then $\sigma(p) = 1 + p$ and $\varphi(p) = p - 1$, so

$$f(p) = \frac{(1+p)(p-1)}{p^2} = \frac{p^2 - 1}{p^2} = 1 - \frac{1}{p^2}.$$

As p increases, $1/p^2$ decreases, so the sequence

$$f(p_1),\ f(p_2),\ f(p_3), \ldots$$

is strictly increasing. Moreover

$$\lim_{i\to\infty} f(p_i) = \lim_{p\to\infty} \left(1 - \frac{1}{p^2}\right) = 1.$$

Hence condition (2) is satisfied with $t = 1$. □

6.1 Answering Question 1.

We are now able to explain why these 'dense lines' are formed. From **Lemma 5** it can be seen that if p is prime, then

$$f(p) = \frac{p^2 - 1}{p^2} \tag{6.3}$$

Since p^2 is generally much larger than 1, $f(p)$ will often be very close to 1. We might guess that the points which form the 'dense line' at $y = 1$ are points of the form $(p, f(p))$, where p is prime. In **Figure 5** below, points of this form are highlighted, and their pattern seems to confirm that our definition of 'dense lines'.

Theorem 2. *A dense line is a sequence $a_i = a_0 \cdot p_i$, for any positive integer a_0.*

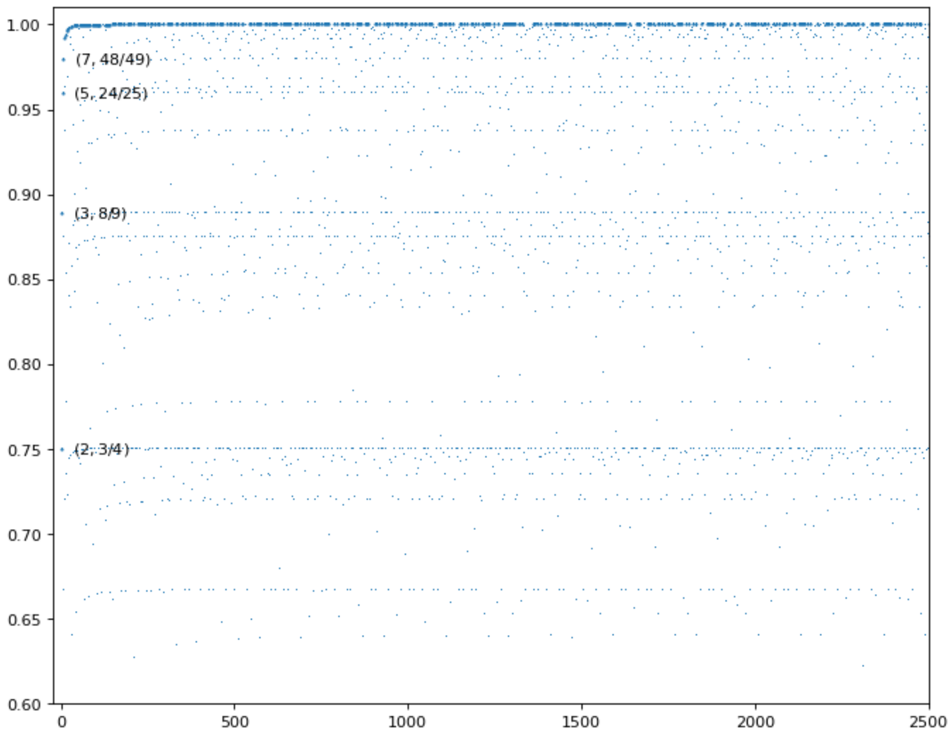

Figure 5: Highlight the points $(p, f(p))$

7 Answering Question 2.

The answer is no, and we present an algorithm which shows this, which relies on Nagura's bound on *Bertrand's Postulate.*

Theorem 3. *For all $n \geq 25$ there is always a prime p with $n < p < 1.2n$*

And a bit of computer brute-force for $3 < n < 25$

and a tiny bit of brute-force by hand for $n = 2, n = 3$

After that though Nagura + induction can get the remaining cases.

8 Conclusion

References

[1] Tom M. Apostol. “A proof that Euler missed: evaluating $\zeta(2)$ the easy way”. In: *The Mathematical Intelligencer* 5 (1983), pp. 59–60. DOI: 10.1007/BF03026576.

[2] Tom M. Apostol. *Introduction to Analytic Number Theory*. New York: Springer-Verlag, 1976.

[3] Grant Sanderson. *Why is pi here? And why is it squared? A geometric answer to the Basel problem*. YouTube video, 3Blue1Brown. Accessed: 2025-10-12. Mar. 2018. URL: https://www.youtube.com/watch?v=d-o3eB9sfls.

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